Yu-Hsuan Wu

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Education

Sept 2020 – June 2024 **National Taiwan University**, BBA in Information Management – Taipei, Taiwan

Research Interests

I am interested in how programming languages behave in the early stages of quantum computer development, with the solid goal of developing theories and tools that facilitate the reliability of quantum program.

Keywords: programming languages, program verification, quantum program

Work Experience

Aug 2024 – present **Full-Time Research Assistant**, Institute of Information Science, Academia Sinica – Taipei, Taiwan
- Supervisor: Prof. Yu-Fang Chen
- On leave during Dec 2024 – Apr 2025 for obligatory military service in the R.O.C. (Taiwan) Army.

July 2023 – Aug 2023 **Summer Research Intern**, Institute of Information Science, Academia Sinica – Taipei, Taiwan
- Supervisor: Prof. Shin-Cheng Mu

Research Projects

Apr 2025 – present **PulseDSL – A Pulse-Level Domain Specific Language for Quantum Control**
Collaborator: Prof. Yuxiang Peng, Purdue University
- Designed the syntax and semantics of the language, and implemented in C language.
- Provided additional APIs for further usage in Python.

Sept 2024 – present **Decision Diagram-Based Quantum Program Verification Tool**
Supervisor: Prof. Yu-Fang Chen, Academia Sinica
- Utilized weighted binary decision diagram to reduce the structural alternations between operations, and designed a minimization algorithm to reduce the size of the diagram from linear growth to constant.

July 2023 – Aug 2023 **Haskell-Based Parser for Markdown-like Language**
Supervisor: Prof. Shin-Cheng Mu, Academia Sinica
- Employed monad transformers for auto-indexing functionality.

Teaching Experience

Fall 2024 **Teaching Assistant - IM2009 Algorithms**, National Taiwan University

Summer 2024 **Teaching Assistant - EMBA7003 Business Decision Making**, National Taiwan University

Fall 2023, Fall 2022 **Teaching Assistant - IM1003 Programming Design**, National Taiwan University

Spring 2023 **Teaching Assistant - IM2010 Operations Research**, National Taiwan University

Fall 2021 – Spring 2023 **Student Instructor - IM Dept. Calculus Remedial Session**, National Taiwan University

Skills

Programming: Proficient with Python, C/C++, and Git

Languages: English (fluent, TOEFL iBT: 108/120), Taiwanese Mandarin (native)